GG-T-311a
SEPTEMBER 2, 1953
SUPERSEDING
Int. Fed. Spec.
GG-T-00311 (GSA-FSS)
September 18, 1952 and
Fed. Spec. GG-T-311
October 2, 1933

FEDERAL SPECIFICATION

THERMOMETERS, SELF-INDICATING, MERCURY IN GLASS, (FEVER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers fever self-indicating, mercury-in-glass thermometers of the types commonly used for measuring body temperatures, except that special thermometers having different ranges and degrees of subdivision designed for a specific use such as for tubercular cases and for ovulation temperature measurements are not included.

1.2 Classification. — This specification covers two types and six classes of fever thermometers.

1.2.1 *Types.*—Fever thermometers covered by this, specification shall have glass-stem lens front and white-enameled backing and shall be of the following types, as specified in the invitation for bids:

Type I. —Flat magnifying lens.
Type II.—Triangular magnifying lens.

1.2.2 *Classes.*—Fever thermometers shall be furnished in the following classes, as specified:

Class A.—Oral bulb, Fahrenheit scale (cylindrical).

Class B.—Oral bulb, centigrade (Celsius) scale (cylindrical).

Class C.—Rectal bulb, Fahrenheit scale (pear shaped).

Class D.—Rectal bulb, centigrade (Celsius) scale (pear shaped).

Class E.—Oral or rectal bulb, Fahrenheit scale (stubby).

Class F.—Oral or rectal bulb, centigrade (Celsius) scale (stubby).

2. APPLICABLE SPECIFICATIONS

2.1 There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal Specifications and the Index of Federal Specifications and Standards may be obtained upon application accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices. Single copies of this specification and other product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago,

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Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Stems.*—The stems shall be of lens front thermometer tubing having white enameled backing. The tubing shall be Corning Normal standard thermometer lead glass, or equal, having a softening point of 648° ± 5°C. and a coefficient of expansion of 85 ± 2 x 10^{-7} centimeters/centimeters/°C.

3.1.2 *Bulbs.*—The bulbs shall be made of Corning Normal glass, or equal, having a softening point of 702° ± 5°C. and a coefficient of expansion of 86 ± 2 x 10^{-7} centimeters/centimeters/°C. Colored bulbs or bulbs blown from the material of the stem shall not be used.

3.1.3 *Mercury.* — The mercury shall be pure, clean, dry, and free from foreign material.

3.1.4 *Pigment.* — The pigment used for filling marks shall be a permanent color complying with the test (see 4.3.2). As a result of the test, the pigment shall not fade or discolor in such a manner as to impair legibility. The loss or absence of pigment from one complete graduation or numeral or the equivalent shall be cause for rejecting the thermometer. The quantity under test may be considered to have passed the permanence of pigment test provided not more than 2 percent of those tested fail.

3.1.5 *Cases.* — Unless otherwise specified, cases shall be made of aluminum, nickel- or chromium-plated metal, hard rubber, or plastic material acceptable to the purchaser. When specified in the invitation for bids, a case shall be furnished with each thermometer delivered. Cases shall be of such size as to be suitable for any of the herein specified thermometers when properly protected by cushioning material inside the case. The length of the case shall not exceed 4¼ inches overall. All cases shall be well constructed.

3.1.6 Magnifying lens tubing.

3.1.6.1 Flat magnifying lens tubing for type I thermometers shall have the numerals and graduations on a flat glass surface or surfaces adjacent to the magnifying lens and perpendicular to the line of sight.

3.1.6.2 Triangular magnifying lens tubing for type II thermometers shall be triangular in cross section with the magnifying lens at the junction of two sides with the numerals and graduations placed on the sides adjacent to the magnifying lens.

3.2 Dimensions and markings.

3.2.1 *Length of thermometers.*—The total length of any thermometer shall be not more than 4⅛ inches nor less than 3⅞ inches.

3.2.2 *Thickness of stem.*

3.2.2.1 *Type I—flat magnifying stem.*—Unless otherwise specified, the cross section of the stem shall be such that no dimension is greater than 0.300 inch and no dimension is less than 0.145 inch.

3.2.2.2 *Type II—triangular magnifying stem.*—The cross section of the stem shall be such that no dimension is greater than 0.235 inch and no dimension is less than 0.145 inch.

3.2.3 *Scale and numerals.*—The scale and numerals shall be clearly etched and the marks filled with pigment. The graduation marks shall be substantially straight, regularly spaced, of uniform width, and perpendicular to the axis of the stem. The width of the graduation marks shall not be greater than 0.012 inch nor less than 0.004 inch, but the width shall in no case be more than the distance between graduation marks. Numerals shall be so placed that the marks to which they pertain can be readily identified.

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3.2.4 *Identification marking.*—Each thermometer shall bear in permanent legible characters, filled with a pigment, the name or registered trade-mark of the manufacturer, and either a serial number or a number and year, to provide complete identification.

3.3 **Aging.** All thermometers shall be aged by either natural or artificial means so that the performance requirements involved will be met after the thermometer has been introduced into commerce. The natural method involves holding the thermometer at room temperature for a period of 4 months after completion of the constriction. The artificial method involves heat treatment of bulb and constriction at a temperature of at least 840°F. for at least 3 hours, followed by cooling to 500°F. at a rate not to exceed 100° per hour.

3.4 **Hard shakers.**—All thermometers shall be so made that the column of mercury can be shaken down by hand without undue effort as determined by the tests specified in 4.3.6.

3.5 **Oral (cylindrical) bulbs.** — Classes A and B.

3.5.1 *Bulb dimensions.*—The diameter of the bulb shall not be less than 0.078 inch nor more than 0.150 inch. The length shall not be less than $\frac{1}{16}$ inch nor more than $\frac{3}{4}$ inch, measurements to be taken from the end of the white-enamel backing.

3.6 **Rectal (pear shaped) bulbs.**—Classes C and D.

3.6.1 *Bulb dimensions.*—The smaller end of the bulb shall be joined to the stem. The greatest diameter shall be not less than 0.15 inch nor more than 0.24 inch. The length shall be not less than $\frac{1}{4}$ inch nor more than $\frac{1}{2}$ inch, measurements to be taken from the end of the white-enamel backing.

3.7 **Oral or rectal (stubby) bulbs.—Classes E and F.**

3.7.1 *Bulb dimensions.*—The diameter of the bulb shall be not less than 0.148 inch nor more than 0.230 inch. The length shall be not less than 0.125 inch nor more than 0.281 inch, measurements to be taken from the end of the white-enamel backing.

3.8 **Fahrenheit thermometers.**—Classes A, C, and E.

3.8.1 *Range of scale.*—The scale shall cover a range of at least 96° to 106°F., but may extend below 96°F. or above 106°F., or both. The 96°F. mark shall be not less than $\frac{3}{8}$ inch from the top of the constriction, i.e., from the point at which the capillary resumes its normal shape above the constriction. The 106°F. mark shall be at least $\frac{3}{8}$ inch from the end of the thermometer or $\frac{1}{8}$ inch below any part of any cap attached to the thermometer, and at least $\frac{1}{8}$ inch from the end of the bore.

3.8.2 *Graduations.*—Thermometers shall be graduated in intervals of 0.2°F.; each degree mark and the mark at the normal point (98.6°) shall be longer than the intervening marks.

3.8.3 *Numerals.*—Numerals shall be etched on the scale at even-numbered degree marks.

3.8.4 *Normal point.*—The graduation at 98.6°F. shall be designated by an arrow or other suitable mark within a tolerance of plus or minus $\frac{1}{2}$ scale division.

3.8.5 *Degrees per inch of scale.* — There shall be not more than 9.0°F. per inch of scale.

3.8.6 *Accuracy.* — At the 98° and 102°F. points, a thermometer shall be accurate within 0.20°F., and at the 106° point, within 0.30°F.

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3.8.7 *Consistency of reading.* — The two readings of the thermometer at 98° and 102°F. shall not differ from each other by more than 0.20°F., and the two readings at 106°F. shall not differ from each other by more than 0.30°F.

3.9 **Centigrade (Celsius) thermometers.**— Classes B, D, and F.

3.9.1 *Range of scale.*—The scale shall cover a range at least from 35.5° to 41.0°C., but may extend below 35.5°C. or above 41.0°C., or both. The 35.5°C. mark shall be not less than ⅝ inch from the top of the constriction, i.e., from the point at which the capillary resumes its normal shape above the constriction. The 41.0°C. mark shall be at least ⅝ inch from the end of the thermometer or ⅛ inch below any part of any cap attached to the thermometer, or at least ⅛ inch from the end of the bore.

3.9.2 *Graduations.*—Thermometers shall be graduated in intervals of 0.1°C., each degree and half-degree mark shall be longer than the intervening marks.

3.9.3 *Numerals.*—Numerals shall be etched on the scale at each degree mark, except at 37°.

3.9.4 *Normal point.* — The graduation at 37°C. shall be designated by an arrow or other suitable mark within a tolerance of plus or minus ½ scale division.

3.9.5 *Degree per inch of scale.*—There shall be not more than 5.0°C. per inch of scale.

3.9.6 *Accuracy.* — At the 37° and 39°C. points, the thermometer shall be accurate within 0.11°C., and at the 41°C. point, within 0.17°C.

3.9.7 *Consistency of reading.* — The two readings of a thermometer at 37° and 39°C. test points shall not differ from each other by more than 0.11°C., and the two readings at 41°C. shall not differ from each other by more than 0.17°C.

3.10 **Entrapped gas.**—Thermometers determined to contain gas when tested in accordance with paragraph 4.8.5 shall be considered defective.

3.11 **Workmanship.** — Workmanship shall be of the best commercial grade. All thermometers shall be free from constructional defects which prevent observations of temperatures within a tolerance of plus or minus 0.2°F. or plus or minus 0.1°C. in the range of 96° to 104°F. or 35° to 40°C., and within plus or minus 0.3°F. or plus or minus 0.15°C. in the range of 104° to 106°F., or 40° to 41°C. Unhealed fire cracks or fractures shall be considered evidence of discreditable workmanship.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Methods of sampling and test.**—Thermometers purchased under this specification may be tested either by testing each thermometer delivered (complete test) or by utilizing a sampling plan specified in 4.1.2, unless otherwise specified in the invitation for bids and contracts for purchase.

4.1.1 *Complete test.*—When each thermometer delivered is to be tested, and test of any portion as large as 10 percent of the whole number shows that more than 10 percent of the thermometers tested fail to meet the major requirements of this specification (3.4, 3.8.6, 3.8.7, 3.9.6, 3.9.7, 3.10, 3.11), or more than 6 percent fail to meet the pigment test, such failure shall be considered cause for rejection of the entire delivery. When complete test of a delivery is made, only those thermometers meeting the requirements of this specification shall be accepted, and the contractor shall replace those failing to meet the requirements with an equal number which do meet the requirements, as specified.

4.1.2 *Sampling plans (for all defects other than defective pigment).* — Acceptance or rejection of the thermometers shall be deter-

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mined by means of a single sampling plan or a sequential sampling plan as specified in the invitation for bids and contracts for purchase. In the use of these plans, major and minor defects shall be defined as follows:

Major	Minor
Error in accuracy greater than specified Inconsistent readings greater than specified Gas in column or bulb Hard shaker Crack in stem or bulb	Dimensional requirements Identification requirements Poor workmanship not resulting in a major defect

If the thermometer has more than one type of defect, each type of defect (major or minor) shall be tallied once, unless a particular type of defect occurs more than once in the same thermometer, or one defect is unquestionably caused by another defect which has already been tallied. In the latter two cases, the principal defect only shall be tallied. Concurrent with testing, utilizing the sampling plan a separate sampling plan shall apply for testing of permanency of pigment (see 4.1.2.3). Failure to meet the requirements of specified sampling plan or pigment sampling plan shall be cause for rejection of the lot involved.

4.1.2.1 *Single sampling plan.* — The single sampling plan shall be as follows:

Lot size	Sample size	Acceptance number	
		Major defects	Minor defects
500- 1,000	100	3	4
1,001- 2,000	200	7	9
2,001- 5,000	300	10	13
5,001-10,000	400	12	17
10,001-25,000	500	15	22
25,001-50,000	1000	30	45

A lot shall be rejected if the number of defects found in the sample exceeds the acceptance number either for major or minor defects.

4.1.2.2 *Sequential sampling plan.* — The sequential sampling plan shall be as follows:

Lot Size	Sample Size Sequential	Cumulative sample size	Major defects		Minor defects	
			Accept	Reject	Accept	Reject
Over 10,000	100	100	0	7	1	8
	100	200	4	9	5	13
	100	300	6	13	10	18
	100	400	9	16	14	22
	100	500	12	19	18	26
	100	600	15	22	23	31
	100	700	19	25	27	35
	100	800	24	25	34	35

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The sequential sampling plan is recommended for lot sizes greater than 10,000 thermometers. In using this plan, the first sample of 100 shall be tested. If both the number of major defects and the number of minor defects are equal to or less than their respective acceptance numbers, the lot shall be accepted. If either the number of major defects or the number of minor defects is greater than its respective rejection number, the lot shall be rejected. If the number found lies between the acceptance and rejection numbers, a second sample shall be tested and the procedure repeated, etc. Once a lot has been accepted for one class of defects (major or minor), testing for that class of defects for which the lot has been accepted may be discontinued.

4.1.2.3 *Sampling plan, defective pigment.*— The sampling plan for testing permanence of pigment shall be a single sampling plan, as follows:

Lot Size	Sample Size		Test "A" Acceptance Number	Test "B" Acceptance Number
	Test "A"	Test "B"		
500-10,000	50	50	1	1
Over 10,000	100	100	2	2

Test "A" and "B" shall be performed as specified in 4.3.2. If, in a given lot of thermometers, more than one batch of pigment is employed, samples shall be taken from each batch in lieu of each lot. Once a lot has been accepted for permanence of pigment, testing for pigment permanence in that lot may be discontinued.

4.2 Inspection.—Samples shall be inspected for material, dimensions and markings, aging, shaking, accuracy, consistency of reading, entrapped gas, workmanship, packaging, packing, and marking.

4.3 Tests. — Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification. In the absence of detailed descriptions of test methods in these specifications, compliance with the requirements shall be determined by methods appropriate to the several requirements.

4.3.1 *Aging.* — All thermometers shall be aged by either natural or artificial means so that the performance requirements involved will be met after the thermometers are introduced into commerce. The natural method involves holding at room temperature for a period of 4 months after completion of the constriction. The artificial method involves heat treatment of the bulb and constriction at a temperature of at least 840°F. for at least 8 hours, followed by cooling to 500°F. at a rate not to exceed 100°F. per hour. In the absence of evidence satisfactory to the purchasing officer that the requirements outlined above have been met, the thermometers may be held for a period of not more than 3 months after delivery before testing.

4.3.2 *Test for permanence of color markings.* — Sample thermometers as specified under 4.1.2.3 shall be subjected to the following tests. During the tests, the solutions shall be maintained at a temperature between 70° and 90°F. Removal periods shall be approximately of 15 minutes duration, or as long as required to dry a thermometer. The test schedule shall be of alternate immersion in the solution specified for periods of one hour and drying the thermometers by wiping with soft tissue or a turkish towel. Five such complete schedules of immersion and drying shall be completed on each of two successive days. The thermometers shall remain in the solution in the intervening night. At the conclusion of each test, the markings on each thermometer shall meet the specified requirements in 3.1.4.

4.3.2.1 *Test "A".* — Sample thermometers shall be totally immersed in a solution consisting of one part of tincture of green soap, N.F., and four parts of water.

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4.3.2.2 *Test "B".* — Sample thermometers shall be totally immersed in a 5 percent by weight phenol solution.

4.3.3 *Tests for thickness of stem.* — Stem dimensions shall be tested with ring gages of the maximum sizes specified and a slot gage on the minimum size specified.

4.3.4 *Tests for diameter of bulb.* — The diameter of the bulbs shall be tested with maximum and minimum ring gages of the sizes specified.

4.3.5 *Test for entrapped gas.*

4.3.5.1 Thermometers in which visual inspection shows the presence of gas in the bulb shall be classified as defective.

4.3.5.2 All thermometers in which there is sufficient gas or other foreign material to cause separation of the mercury column above the constriction during shipment, test, or use, shall be classified as defective. To test for this condition, the thermometer shall be heated so that the reading is 102° F. or 39°C. The mercury shall be shaken away from the bulb in the centrifuge at 200 r.p.m. with the top of the thermometer 32 cm. from the axis of rotation. A broken mercury column above the constriction shall be an indication that there is gas or other foreign material present.

4.3.6 *Test for hard shakers.*

4.3.6.1 *All thermometers*, after having been heated to 106°F. or 41°C., shall be mounted in a centrifuge, and whirled at the specified speed. For a centrifuge in which the ends of the bulbs are 17 cm. from the axis of rotation, the specified speed is 580 r.p.m. For a centrifuge in which the ends of the bulbs are 32 cm. from the axis of rotation, the specified speed is 400 r.p.m. The larger type is preferred. When whirled at the specified speed, the index shall fall below 96°F. or 35°C. Thermometers which do not comply shall be classified as defective.

4.3.7 *Tests for accuracy and consistency of reading.*

4.3.7.1 Thermometers shall be compared at 98°, 102°, and 106°F., and 37°, 39°, and 41°C., with a thermometer certified by the National Bureau of Standards for this purpose by heating in a well-stirred water bath, removing from bath, and reading. The temperature of the bath shall be raised at a substantially uniform rate during the last two minutes before the test temperature is reached, and the rise during this two-minute period shall be not more than 0.2°F. The thermometer shall be removed from the bath as soon as the temperature of test is reached, except in the second test at 106°F. where the procedure specified under paragraph 4.3.7.2 shall be followed. The reading shall be observed after removal from the bath. A suitable magnifying glass shall be used and the readings estimated to the nearest one-tenth division (graduation interval), or better. Two independent comparisons shall be made and recorded on each thermometer at each test temperature.

4.3.7.2 In the first test, at 106°F., or 41°C., the thermometers shall be removed from the bath as soon as the test temperature is reached. In the second test, at 106°F. or 41°C., the bath shall be brought up to the test temperature and then allowed to cool slowly to 105°F. or 40.4°C., or below, at a uniform rate not exceeding 1°F. or 0.55°C. in three minutes. The thermometers shall then be removed from the bath and read, and the readings recorded as before.

4.3.7.3 *Accuracy.*—No individual reading, in comparison with the certified standard, shall be in error by more than the amount permitted in 3.8.6 or 3.9.6.

4.3.7.4 *Consistency.*—The two readings at each test point shall not differ from each other by more than the amount permitted in 3.8.7 or 3.9.7, except as provided in 4.3.7.5.

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4.3.7.5 If either reading at one and only one test temperature is in error by more than the tolerance specified in 4.3.7.3, or if the readings at one and only one test temperature differ from each other by more than the tolerance specified in 4.3.7.4, the thermometer shall be completely retested for consistency of reading. If, on retest, the thermometer fails to meet the requirements of 4.3.7.3 or 4.3.7.4, it shall be classified as defective.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**—When thermometers are to be supplied with cases as specified in 3.1.5, each thermometer shall be packaged in its case for shipment, or the cases and thermometers shall be packaged separately, as specified by the purchaser. When the thermometers are supplied in cases, commercial packages, unless otherwise specified, are acceptable under this specification. When thermometers are supplied without cases, or are packaged separately from the cases, they shall be securely wrapped in bundles of not less than 6 nor more than 40, and the bundle protected and packaged in such manner as to insure arrival in good condition at destination.

5.2 **Packing.** — The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.**

5.3.1 *Packages.*—Unless otherwise specified, packages shall be marked with the name and size of the commodity, the quantity contained therein, and the names of the manufacturer and contractor.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name and size of the commodity as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Purchasers should exercise all options offered herein (1.2.1, 1.2.2, 3.1.5, 3.2.2.1, 5.1, 5.3 and sec. 7) and should specify, both in the invitation for bids and in purchase order, whether acceptance is to be based on complete tests (see 4.1.1) or test of representative samples (see 4.1.2).

6.2 Purchasers should specify whether thermometers are to be supplied with or without cases.

6.3 These specifications do not cover fever thermometers for veterinary use.

6.4 This specification covers the following items appearing in the Armed Services Catalog of Medical Materiél:

7–931–050 Thermometer, Clinical, Human, Oral, 6s.
7–931–060 Thermometer, Clinical, Human, Rectal, 6s.
7–932–000 Thermometers, Clinical, Oral: Fahrenheit scale.
7–932–075 Thermometers, Clinical, Rectal: Fahrenheit scale.

6.5 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

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6.6 This specification governs all United States Government purchases of this commodity. Users are requested to retain this specification in permanent file until a revision is issued.

6.7 The specification for the commercial purchase of clinical thermometers currently approved is Commercial Standard CS1–52, in respect to requirements of quality and performance, and in the tests to be used in determining compliance with those requirements. In addition to the requirements of the commercial standard, this specification contains requirements as to type, form, size and pigment test, which are necessary to describe definitely the instruments which the Government wishes to purchase.

6.8 Persons furnishing thermometers under these specifications should be prepared to furnish evidence, satisfactory to the purchasing officer, that the thermometers will meet the aging requirements specified in 4.3.1.

6.9 Attention of persons furnishing thermometers under these specifications, when test is to be made by sampling, is called to the fact that in order to meet the requirements herein stated in the sample, it will probably be necessary to adhere to closer tolerances in making their own tests than are provided in these specifications.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments:

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN–B–621—Boxes; Wood, Nailed and Lock-Corner.

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

Military Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

JAN–P–120—Packaging and Packing for Overseas Shipment — Cartons, Folding, Paperboard.

JAN–P–133—Packaging and Packing for Overseas Shipment — Boxes, Set-Up, Paperboard.

MIL–L–10547—Liners, Case, Waterproof.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

7.2.2 *Inspection procedures (Navy only).*—The general inspection procedures shall be

[1] Applicable to Navy purchases only.

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in accordance with General Specifications for Inspection of Material.

7.2.3 *Packaging for domestic and overseas shipment.*

7.2.3.1 *Hospital pack.*—Six thermometers without cases of the stock number specified constitute one unit.

7.2.3.2 *Individual pack.*—One thermometer packaged in an individual case as specified constitutes one unit.

7.2.3.3 *Unit packaging.*

7.2.3.3.1 *Hospital pack.*—Six thermometers (1 unit) shall be packaged in a carton of appropriate size constructed in accordance with the requirements of Military Specification JAN-P-120 or JAN-P-133. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to individually protect each thermometer from damage.

7.2.3.3.2 *Individual pack.*—Unit packaging not required.

7.2.3.4 *Intermediate packaging.*

7.2.3.4.1 *Hospital pack.*—Twenty-four unit cartons (24 units) of one stock number shall be packaged in a carton of appropriate size constructed in accordance with the requirements of Federal Specification LLL-B-631. Each carton shall be adequately secured to prevent accidental opening.

7.2.3.4.2 *Individual pack.* — Twelve thermometers (12 units) of one stock number shall be packaged in a carton of appropriate size constructed in accordance with the requirements of Military Specification JAN-P-120 or JAN-P-133. Each carton shall be adequately secured to prevent accidental opening.

7.2.4 *Packing.*

7.2.4.1 *For domestic shipment.*—The number of intermediate cartons specified in table I shall be overpacked in an exterior container designed for type 8 load, and constructed in accordance with the requirements of Federal Specification NN-B-621. Boxes shall be strapped when required by appendix thereto.

TABLE I.—*Packing quantities*

Stock No.	Type of pack	Number of intermediate cartons to be overpacked in each shipping container	Total number of units to be contained in each shipping container
7-931-050	Hospital	12	288
7-931-060	Hospital	12	288
7-932-000	Individual	144	1728
7-932-075	Individual	144	1728

7.2.4.2 *For overseas shipment.*—The number of intermediate cartons specified in table I shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with the requirements of Military Specification JAN-P-106. Closure and strapping shall comply with appendix thereto.

7.2.4.2.1 *Case liner.* — Each box shall be lined with a waterproof case liner conforming to the requirements of Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

7.2.4.3 *Packing variation permitted.* — If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in an exterior shipping container of suitable size

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and design and so constructed as to insure safe delivery to specified destination.

7.2.5 *Marking.* — Each unit carton, each intermediate carton and each shipping container shall be marked in accordance with Military Standard MIL-STD-129.

7.2.6 *Ordering data.*—Procurement documents should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.4.1 and 7.2.4.2.)

ARMY INTEREST: M
NAVY INTEREST: Md

This specification is a part of Section IV, Part 6, of the Federal Standard Stock Catalog.

☆ U. S. GOVERNMENT PRINTING OFFICE: 1963

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or Contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY (Printed or typed name and activity)	DATE

DD, FORM 1426 REPLACES NAVSHIPS [illegible]

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
DEFENSE PERSONNEL SUPPORT CENTER (MEDICAL)
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

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PLATE NO. 15410 (

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